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Descent for

idempotent algebras

Last time Convolution products on $\mathbb{R}$ -vector spaces, the relative homology functor

$$\Gamma_{M_{\mathbb{R}}}(\mathrm{LCH}/_{M_{\mathbb{R}}}, \otimes) \longrightarrow (\mathrm{Sh}(M_{\mathbb{R}}), *)$$

the symmetric monoidal left adjoints

$$\Psi_{\sigma}(\mathrm{Fun}(\Theta(\sigma)^{\mathrm{op}}, \mathrm{Sp}), \otimes_{\mathrm{Day}}) \longrightarrow (\mathrm{Mod}_{\omega_{\sigma}} \mathrm{Sh}(M_{\mathbb{R}}), *)$$

and the lax symmetric monoidal coherent-constructible correspondence

$$k_{\Sigma}(\mathrm{QCoh}(X_{\Sigma}/T), \otimes) \longrightarrow (\mathrm{Sh}(M_{\mathbb{R}}), *)$$

Today Descent for idempotent algebras & symmetric monoidality of k_{Σ}

Recall. For a lattice N and cone $\sigma \in N_{\mathbb{R}}$, we considered

$$\Psi_{\sigma}: \Theta(\sigma) \hookrightarrow \text{Mod}_{\omega_{\sigma}} \text{Poly}(M) \xrightarrow{\Gamma_{M|R}} \text{Mod}_{\omega_{\sigma}} \text{Sh}(M_{|R})$$

$$X \longmapsto \omega_X.$$

using the universal property of Day convolution, we extended this to define Ψ_{σ} .

> Previously, for a fan Σ in N , we constructed an equivalence

$$\lim_{\sigma \in \Sigma^{\circ}} \text{Fun}(\Theta(\sigma)^{\text{op}}, \text{Sp}) \xrightarrow{\sim} \text{QCoh}(X_{\Sigma}/T).$$

> Taking limits, we get a sym. mon. functor

$$\Psi_{\Sigma}: \lim_{\sigma \in \Sigma^{\circ}} \text{Fun}(\Theta(\sigma)^{\text{op}}, \text{Sp}) \longrightarrow \lim_{\sigma \in \Sigma^{\circ}} \text{Mod}_{\omega_{\sigma}} \text{Sh}(M_{|R})$$

> Since each inclusion $\text{Mod}_{\omega_{\sigma}} \text{Sh}(M_{|R}) \hookrightarrow \text{Sh}(M_{|R})$ is lax sym. mon. and both a left and a right adjoint, the induced functor

$$R_{\Sigma}: \lim_{\sigma \in \Sigma^{\text{op}}} \text{Mod}_{\omega_{\sigma}} \text{Sh}(M_R) \longrightarrow \text{Sh}(M_R)$$

is lax sym. mon. and both a left and right adjoint

- > The coherent-constructible correspondence is the composite lax sym. mon. left adjoint

$$\text{QCoh}(X_{\Sigma}/T) \simeq \lim_{\sigma \in \Sigma^{\text{op}}} \text{Fun}(\mathcal{O}(\sigma)^{\text{op}}, \text{Sp}) \xrightarrow{R_{\Sigma} \circ \Psi_{\Sigma}} \text{Sh}(M_R)$$

- > First, we want to see that, at least when Z is smooth projective, R_{Σ} is an equivalence

Descent for idempotent algebras

Lemma. Let $\mathcal{C}$ be a presentably sym. mon. ∞ -category. Then

(1) The poset $\text{CAlg}^{\text{idem}}(\mathcal{C})$ has all joins (= unions = coproducts).

Binary joins are given by

$$A \vee B = A \otimes B,$$

and for a small set I , we have

$$\bigvee_{i \in I} A_i = \operatorname{Colim}_{\substack{J \subset I \\ \text{finite}}} \bigotimes_{j \in J} A_j.$$

(2) The poset $\text{CAlg}^{\text{idem}}(\mathcal{C})$ has all meets (= intersections = products).

Binary meets are given by

$$A \wedge B = A \times_{A \otimes B} B,$$

$$A \simeq A \otimes 1 \xrightarrow{\text{id} \otimes \text{unit}} A \otimes B$$

$$B \simeq 1 \otimes B \xrightarrow{\text{unit} \otimes \text{id}} A \otimes B$$

and for a small set I ,

$$\bigwedge_{i \in I} A_i = \lim_{\substack{\emptyset \neq J \subset I \\ \text{finite}}} \bigotimes_{j \in J} A_j.$$

Proposition Let $\mathcal{C}$ be a presentably Sym. mon. ∞ -category. The ∞ -category $\mathrm{CAlg}^{\mathrm{idem}}(\mathcal{C})^{\mathrm{op}}$ has a Grothendieck topology called the coherent topology where:

(*) A family of maps $(f_i: A \rightarrow A_i)_{i \in I}$ is covering if and only if there is a finite subset $I_0 \subset I$ such that the maps $(f_i)_{i \in I_0}$ exhibit A as the meet $\bigwedge_{i \in I_0} A_i$.

Proposition Let $\mathcal{C}$ be a presentably symmetric monoidal ∞ -category. Then the functor

$$\mathrm{Mod}_{(-)}(\mathcal{C}): \mathrm{CAlg}^{\mathrm{idem}}(\mathcal{C}) \longrightarrow \mathrm{CAlg}(\mathrm{Pr}^{\mathrm{L}})$$

with functoriality extension of scalars is a sheaf for the coherent topology.

Example Let X be a scheme (in the ordinary sense, but one can also generalize this to stacks over $\mathcal{S}$) and let $j: U \hookrightarrow X$ be a quasicompact open. Then $j_*: \mathrm{QCoh}(U) \rightarrow \mathrm{QCoh}(X)$ is fully faithful and admits both a left and right adjoint. Moreover, the quasicoherent

Sheaf $j_*(\mathcal{O}_U)$ is an idempotent algebra in $\text{QCoh}(X)$ and

$$\text{QCoh}(U) \simeq \text{Mod}_{j_*(\mathcal{O}_U)} \text{QCoh}(X),$$

This comes from the projection formula: for a qcqs map $f: Y \rightarrow X$, there is a natural equivalence

$$f_*(F) \otimes_{\mathcal{O}_X} G \xrightarrow{\sim} f_*(F \otimes_{\mathcal{O}_Y} f^*(G))$$

Hence

$$j_*(\mathcal{O}_U) \otimes_{\mathcal{O}_X} j_*(\mathcal{O}_U) \xrightarrow{\sim} j_*(\mathcal{O}_U \otimes_{\mathcal{O}_U} \underbrace{j^* j_*(\mathcal{O}_U)}_{\simeq \mathcal{O}_U \text{ since } j_* \text{ is fully faithful}}) \simeq j_*(\mathcal{O}_U)$$

If $(U_i)_{i \in I}$ is a finite family of quasi-compact opens that cover X , one can show that $(\mathcal{O}_X \rightarrow j_{i,*}(\mathcal{O}_{U_i}))_{i \in I}$ form a covering for the coherent topology. In this case, descent recovers Zarski descent for finite coverings.

Example. Let X be a topological space and $i: Z \hookrightarrow X$ a closed subspace. Then $i_*(1_Z)$ is an idempotent algebra in $\mathrm{Sh}(X)$ and identifies $\mathrm{Sh}(Z)$ with $\mathrm{Mod}_{i_*(1_Z)} \mathrm{Sh}(X)$. This follows from the projection formula for closed immersions (more generally, proper maps):

$$i_*(F) \otimes G \xrightarrow{\sim} i_*(F \otimes i^*(G))$$

In this case, if $(Z_j)_{j \in J}$ are finitely many closed subspaces that cover X , one can show that $(1_X \rightarrow i_*(1_{Z_j}))_{j \in J}$ form a cover for the coherent topology. In this case, descent for idempotent algebras says that the diagram

$$\mathrm{Sh}(X) \longrightarrow \prod_{j_0 \in J} \mathrm{Sh}(Z_{j_0}) \rightrightarrows \prod_{j_0, j_1 \in J} \mathrm{Sh}(Z_{j_0} \cap Z_{j_1}) \rightrightarrows \dots$$

is a limit diagram. This descent property for sheaves is probably unfamiliar, but can be proven via other means. See [arXiv:2108.01924, §2.4].

Gluing the idempotent algebras $\omega_{\sigma^\vee}$

> We'll now show that for a smooth projective fan (N, Σ) , the functor R_Σ is fully faithful.

Observation. Let N be a lattice and $\sigma, \tau \in N_{\mathbb{R}}$ cones. Then from our earlier discussion, we have

$$\sigma^\vee + \tau^\vee = (\sigma \cap \tau)^\vee$$

Since the assignment $\sigma \mapsto \omega_{\sigma^\vee}$ sends Minkowski sums to convolution products, we deduce that

$$\omega_\sigma * \omega_\tau \stackrel{\sim}{=} \omega_{(\sigma \cap \tau)^\vee}$$

Notation. Let (N, Σ) be a fan. Write $\Sigma_{\text{top}} \subset \Sigma$ for the subset of top-dimensional cones.

Notation. For a set I , write $\text{Sub}_{\neq \emptyset}(I)$ for the poset of nonempty subsets of I .

Observe. For fan (N, Σ) , there is an adjunction

$$L: \text{Sub}_{\neq \emptyset}(\Sigma_{\text{top}}) \rightleftarrows \Sigma^{\text{op}} : R$$

given by

$$L(S) = \bigcap_{\sigma \in S} \sigma \quad \text{and} \quad R(\tau) := \{\sigma \in \Sigma_{\text{top}} \mid \tau \leq \sigma\}$$

To see that this is an adjunction, note that

$$\tau \leq \bigcap_{\sigma \in S} \sigma \quad \text{if and only if} \quad S \subseteq \{\sigma \in \Sigma_{\text{top}} \mid \tau \leq \sigma\}$$

In particular, since L is a left adjoint, L is a lim -equivalence

Cor. If (N, Σ) is a fan, there is a natural equivalence

$$\lim_{\sigma \in \Sigma^{\text{op}}} \omega_{\sigma v} \xrightarrow{\sim} \lim_{S \in \text{Sub}_{\neq \emptyset}(\Sigma_{\text{top}})} \bigstar_{\tau \in S} \omega_{\tau v}$$

Proof.

Since L is a lim -equivalence, we deduce that

$$\lim_{\sigma \in \Sigma^{\text{op}}} \omega_{\sigma^\vee} \xrightarrow{\sim} \lim_{S \in \text{Sub}_{\neq \emptyset}(\Sigma_{\text{top}})} \omega_{L(S)^\vee}$$

Since $L(S) = \bigcap_{z \in S} z$, we deduce that $L(S)^\vee = \sum_{z \in S} z^\vee$. So

$$\omega_{L(S)^\vee} \simeq \bigstar_{z \in S} \omega_{z^\vee}$$

Plugging this into the above completes the proof. $\square$

Prop. Let (N, Σ) be a smooth projective fan. Then the maps of idempotent algebras for convolution

$$(\mathbb{1}_{\text{conv}} \longrightarrow \omega_{\sigma^\vee})_{\sigma \in \Sigma_{\text{top}}}$$

form a cover of the unit in $(\text{Sh}(M_{\mathbb{R}}), \star)$.

$$\mathbb{1}_{\text{conv}} \longrightarrow \lim_{\sigma \in \Sigma^{\text{op}}} \omega_{\sigma^\vee} \longrightarrow \lim_{S \in \text{Sub}_{\neq \emptyset}(\Sigma_{\text{top}})} \bigstar_{z \in S} \omega_{z^\vee}$$

are equivalences.

Cor. If (N, Σ) is a smooth projective fan, then the lax sym. monoidal functor

$$R_{\Sigma} : \lim_{\sigma \in \Sigma^{\circ}} \text{Mod}_{\omega_{\sigma}} \text{Sh}(M_{\mathbb{R}}) \longrightarrow \text{Sh}(M_{\mathbb{R}})$$

is an equivalence. Hence

$$K_{\Sigma} : \text{QCoh}(X_{\Sigma}/T) \longrightarrow \text{Sh}(M_{\mathbb{R}})$$

is symmetric monoidal.

Recall. The unit for convolution is the skyscraper sheaf

$$O_*(S) = O_!(S)$$

Proof of descent result

We already know that the right-hand map is an equivalence, so it suffices to treat the left-hand map. We show that this map is an equivalence by computing stalks.

> Note that $O_*(S)$ has stalk S at the origin and 0 everywhere else.

> Let $n = \dim(M_{\mathbb{R}})$ and recall that $\omega_{\sigma^\vee} \cong \omega_{\sigma^\vee, 0}$ as sheaves on $M_{\mathbb{R}}$, where $\sigma^\vee_0$ denotes the interior of $\sigma^\vee$.

In particular, if $\sigma \neq 0$, then $(\omega_{\sigma^\vee})_0 = 0$, and if $\sigma = 0$, then $(\omega_{\sigma^\vee})_0 \cong S[n]$.

Stalk at the origin. Since the limit over Σ^{op} is finite, the stalk commutes past this limit. Hence

$$\left(\lim_{\sigma \in \Sigma^{\text{op}}} \omega_{\sigma^\vee} \right)_0 \cong \lim_{\sigma \in \Sigma^{\text{op}}} S_{\{0\}}[n]$$

Here, $S_{\{0\}}: \Sigma^p \rightarrow S_p$ takes the origin to S and is 0 elsewhere.

Note that there is a fiber sequence in $\text{Fun}(\Sigma^p, S_p)$

$$S_{\{0\}} \longrightarrow \underline{S} \longrightarrow S_{\Sigma^p - \{0\}},$$

where $\underline{S}$ is the constant presheaf and $S_{\Sigma^p - \{0\}}$ is the right Kan extension of the constant presheaf on $\Sigma^p - \{0\}$. Taking limits gives the fiber sequence

$$\lim_{\sigma \in \Sigma^p} S_{\{0\}} \longrightarrow S \longrightarrow \lim_{\sigma \in \Sigma^p} S_{\Sigma^p - \{0\}}$$

Since $S_{\Sigma^p - \{0\}}$ is right Kan-extended from $\Sigma^p - \{0\}$, we have

$$\lim_{\sigma \in \Sigma^p} S_{\Sigma^p - \{0\}} \simeq \lim_{\sigma \in \Sigma^p - \{0\}} \underline{S}.$$

Combinatorial Exercise. Show that $B(\Sigma^p - \{0\}) \simeq S^{n-1}$.

For any ∞ -category $\mathcal{C}$, the natural map $\mathcal{C} \rightarrow B\mathcal{C}$ is a
 lim-equivalence and colim-equivalence. In particular,

$$\begin{aligned} \lim_{\sigma \in \Sigma^p, \{0\}} \underline{S} &\simeq \lim_{S^{n-1}} \underline{S} \\ &\simeq S \oplus S[-(n-1)] \end{aligned}$$

Under this identification, the map $S \rightarrow S \oplus S[-(n-1)]$
 in the fiber sequence is the inclusion of the first factor.
 Hence

$$\begin{aligned} \lim_{\sigma \in \Sigma^p} S_{\{0\}} &\simeq \text{fib}(\text{inc. } S \rightarrow S \oplus S[-(n-1)]) \\ &\simeq S[-(n-1)][-1] \\ &\simeq S[-n] \end{aligned}$$

Hence

$$\left(\lim_{\sigma \in \Sigma^p} \omega_{\sigma v} \right)_0 \simeq \lim_{\sigma \in \Sigma^p} S_{\{0\}}[n] \simeq S,$$

as desired.

Stalk at $m \in M_{\mathbb{R}} \setminus \{0\}$.

We want to show that the stalk of the limit is zero

we see that

$$\lim_{\sigma \in \Sigma^{\text{op}}} (\omega_{\sigma, \nu})_m \simeq \lim_{\sigma \in \Sigma^{\text{op}}} \mathcal{S}_{m, +}[\eta],$$

where

$$\begin{aligned} \mathcal{S}_{m, +} : \Sigma^{\text{op}} &\longrightarrow Sp \\ \sigma &\longmapsto \begin{cases} \mathcal{S}, & m \in \sigma^{\vee, 0} \\ 0, & \text{else} \end{cases} \end{aligned}$$

Write $\Sigma_{m, -} = \{\sigma \in \Sigma \mid m \in \sigma^{\vee, 0}\}$, and $\mathcal{S}_{m, -}$ for the right Kan extension of the constant presheaf $\Sigma_{m, -}^{\text{op}} \rightarrow Sp$ at $\mathcal{S}$ to Σ^{op} . Then there is a fiber sequence

$$\mathcal{S}_{m, +} \longrightarrow \underline{\mathcal{S}} \longrightarrow \mathcal{S}_{m, -}$$

in $\text{Fun}(\Sigma^{\text{op}}, Sp)$. Thus to prove the claim, it suffices to

Show that the left-hand map induces an isomorphism on limits. As before, since $S_{m,-}$ is right Kan extended from $\Sigma_{m,-}^{\text{op}}$,

$$\lim_{\sigma \in \Sigma^{\text{op}}} S_{m,-} \simeq \lim_{\sigma \in \Sigma_{m,-}^{\text{op}}} \underline{S}$$

Moreover, the map

$$\underline{S} \simeq \lim_{\sigma \in \Sigma^{\text{op}}} \underline{S} \longrightarrow \lim_{\sigma \in \Sigma_{m,-}^{\text{op}}} \underline{S} \simeq \lim_{B(\Sigma_{m,-}^{\text{op}})} \underline{S}$$

$\nearrow$
 Σ^{op} is contractible

is induced by the inclusion $\Sigma_{m,-}^{\text{op}} \subset \Sigma^{\text{op}}$. Thus we are reduced to showing that the poset $\Sigma_{m,-}$ has contractible classifying anima.

> We'll prove this next time!